

Performance requirements

Purpose of Writing:

3D scanning is a process that involves real-time collection, processing, and rendering of large amounts of 3D data, which places high demands on a computer's computing power, graphics processing, and memory bandwidth. Just as a high-performance sports car requires a highway, your scanner also needs a computer with sufficient performance to fully realize its potential.

If the computer's performance is not up to par, you may encounter the following issues:

- ▶ Startup phase: Performance test failure, software fails to start or crashes, preview failure.
- ▶ Scanning phase: Preview lagging, view loss, making it difficult to control the scanning path; low scanning speed; inability to use the fine scanning mode.
- ▶ Data processing phase: Extremely slow post-processing or even software crashes; inability to perform fine processing, resulting in poor data processing results.

This guide aims to help you:

1. Avoid problems in advance: Before purchasing a scanner or a new computer/phone, scientifically assess the compatibility of the devices to ensure your investment is worthwhile and to achieve a smooth scanning experience.
2. Troubleshoot after purchase: If users who have already purchased encounter problems during use, they can consider the "computer/phone configuration" as a key troubleshooting info, providing accurate information for after-sales technical support and accelerating problem resolution. (the configuration could be pasted from the performance test)

For PC-end Scanning

Understanding Core Hardware Specifications: Taking Windows as an Example

- ▶ **CPU (Central Processing Unit):** It can be regarded as the "brain" of a computer, responsible for the calculation and stitching of 3D data. A higher clock speed and more cores mean faster point cloud stitching and model calculation speeds, significantly reducing the time you wait for processing. **It is recommended to use Intel i7 or above series. AMD is not recommended at present due to stability and compatibility reasons.**
- ▶ **GPU (Graphics Processing Unit) and Graphics Card Type:** These two concepts are highly related. The GPU is the core chip of the graphics card, while the graphics card is a complete hardware assembly that integrates the GPU, video memory, etc. It is the "engine" of the 3D scanning experience, mainly responsible for real-time processing and rendering of 3D images. It directly determines the smoothness of the preview screen and the fineness of model rendering.
 - ▶ **Graphics Card Brand and Series:** For 3D scanning software, **NVIDIA graphics cards are usually a must and often the only choice. AMD or Intel graphics cards may not work properly.**

- **Graphics Card Model:** Among NVIDIA graphics cards, the RTX series (such as **RTX 30/40/50 series**) with dedicated RT Cores (Ray Tracing cores) and Tensor Cores (Tensor cores) can provide more powerful graphics processing and computing capabilities.
- **Video Memory (VRAM): At least 6GB or more.** VRAM is the "dedicated memory" of the graphics card, used to store textures and 3D model data. During high-precision scanning, the model data volume is huge. Large VRAM can ensure that all data can be quickly called and processed, avoiding stuttering and rendering errors due to insufficient VRAM. For scanners like Raptor pro/X and Sermoon series that include large-scale cross-line modes, the data volume is even larger, and it is recommended to increase VRAM to **8GB**.
- **RAM (Random Access Memory):** Memory is the "workbench" of the CPU. 3D scanning, especially high-precision scanning, generates massive point cloud data. Sufficient memory ensures that this data can be smoothly loaded and processed. If the memory is insufficient, the software may limit the number of scanning points, preventing fine scanning and even causing errors or crashes during processing. Large-capacity memory ensures the system has sufficient "working space", avoiding the slow hard disk being used as virtual memory due to insufficient memory, which can cause stuttering. Therefore, **for infrared structured light scanners like Ferret and Otter series, 16GB is currently the recommended starting point for a smooth experience. For blue line laser scanners such as Raptor and Sermoon series, which generate more point cloud data during laser point cloud scanning, it is recommended to increase RAM to 32GB and 64GB.**

Platform	Ferret series	Otter series	Raptor	Raptor X / Raptor Pro	Sermoon series
Windows (Recommended configuration)	OS: Windows 10 / 11 (64-bit only) CPU: Intel Core i7 12th Gen or above GPU: NVIDIA GeForce RTX 3060 (or higher) Memory (RAM): 16 GB or more	OS: Windows 10 / 11 (64-bit only) CPU: Intel Core i7 12th Gen or above GPU: NVIDIA GeForce RTX 3060 (or higher) Memory (RAM): 16 GB or more	OS: Windows 10 / 11 (64-bit only) CPU: Intel Core i7 12th Gen or above GPU: NVIDIA GeForce RTX 3060 (6GB VRAM) or higher Memory (RAM): 32 GB or more	OS: Windows 10 / 11 (64-bit only) CPU: Intel Core i7 13 or above GPU: NVIDIA 4060 (8GB VRAM) or higher Memory (RAM): 32 GB or more	OS: Windows 10 / 11 (64-bit only) CPU: Intel Core i7 13 or above GPU: NVIDIA 4060 (8GB VRAM) or higher Memory (RAM): 32 GB or more
Windows (The lowest configuration)	OS: Windows 10 / 11 (64-bit only) CPU: Intel i5-Gen8 Memory (RAM): 8 GB Software version: ≥V2.3.0	OS: Windows 10 / 11 (64-bit only) CPU: Intel i5-Gen8 Memory (RAM): 8 GB Software version: ≥V2.3.0	OS: Windows 10 / 11 (64-bit only) CPU: Intel i5-Gen8 Memory (RAM): 8 GB Software version: ≥V3.3.0	OS: Windows 10/11 (64-bit only) CPU: i7-Gen7 GPU: NVIDIA (6GB VRAM) RAM: 16GB Software version: ≥V3.3.0	OS: Windows 10/11 (64-bit only) CPU: i7-Gen7 GPU: NVIDIA (6GB VRAM) RAM: 16GB Software version: ≥V3.3.0

Platform	Ferret series	Otter series	Raptor	Raptor X / Raptor Pro	Serm seri
macOS (Recommended configuration)	OS: macOS 14 or above CPU: Apple M2/M3/M4 Memory (RAM): 16 GB or more	OS: macOS 14 or above CPU: Apple M2/M3/M4 Memory (RAM): 16 GB or more	OS: macOS 14 or above CPU: Apple M2/M3/M4 Memory (RAM): 32 GB or more	OS: macOS 14 or above CPU: Apple M3/M4 Memory (RAM): 32 GB or more	OS: mac or above CPU: Ap M3/M4 Memory (RAM): 3 or more

Recommended Computer Graphics Card Model

Note: The following recommended results have been verified through testing and will continue to be tested and updated. :)

Manufacturers	Series	Currently supported graphics card models	Recommendation rate
NVIDIA	GeForce 50 series	RTX5080	★★★★★
		RTX5060	★★★★★
	GeForce 40 series	RTX4080super	★★★★
		RTX4060	★★★★
	GeForce 30 series	RTX3080Ti	★★★
		RTX3060	★★★
	GeForce 20 series	RTX2080super	★★
		RTX2060 6GB	★★
		GTX 1660super 4G	★
	GeForce 10 series	GTX 1080 Ti	★
		GTX 1060 3GB	★
Apple		M4	★★★★★
	M series chips		

Manufacturers	Series	Currently supported graphics card models	Recommendation rate
		M3	★★★★
		M2	★★★
		M1	★★

For Wireless Scanning with smart phone scanning

The wireless handles required for using the wireless scanning function

- ▶ **Raptor series & Sermoon S1 & Otter lite basic**, with Scan Bridge (Raptor X comes with a Scan Bridge and no additional purchase is needed) , Purchase link: <https://store.creality.com/products/creality-scan-bridge> ☑ , choose Scan Bridge for Sermoon S1 & Raptor Series.
- ▶ **Otter**, with Scan Bridge, Purchase link: <https://store.creality.com/products/creality-scan-bridge> ☑ , choose Scan Bridge for Otter Series .
- ▶ **Otter Lite**, with Lite Bridge included in the package.
- ▶ **Ferret pro & Ferret**, with wireless box (Ferret pro comes with a wireless box and no additional purchase is needed) Wireless Bridge (Purchase link: <https://store.creality.com/products/upgrade-kit-for-wireless-scanning> ☑)

Recommended Phones' configuration

Note: Google Pixel series are not recommended for use; Samsung phones with Exynos chips, some phones with MediaTek Dimensity or Huawei Kirin chips, and devices running HarmonyOS are not supported.

Ferret/ Otter Series			Raptor series/Sermoon S1
Android	Chip	Qualcomm Snapdragon 8gen1 or above	Qualcomm Snapdragon 8gen3 or above
	Operation System	Android 10.0+	Android 10.0+
	RAM	8GB +	12GB +
iOS	Chip	A15 or higher	A16 or higher
	Operation System	iOS 15+	iOS 15+

Ferret/ Otter Series		Raptor series/Sermoon S1
RAM	6GB +	8GB +

Recommended Phone models' examples:

Platform	Ferret Series	Otter Series	Raptor/Raptor X	Raptor Pro	Sermoon S1
Android	Samsung: S24 Series Xiaomi: 14 Series Redmi: K70 Pro OPPO: Find N3 Vivo: X90 Pro+ Honor: Magic 5	Samsung: S24 Series Xiaomi: 14 Series Redmi: K70 Pro OPPO: Find N3 Vivo: X90 Pro+ Honor: Magic 5	Samsung: S25 Series Xiaomi: 15 Series Redmi: K70 Pro OPPO: Find N3 Vivo: X90 Pro+ Honor: Magic 5	Samsung: S25 Series Xiaomi: 15 Series Redmi: K70 Pro OPPO: Find N3 Vivo: X90 Pro+ Honor: Magic 5	Samsung: S25 Series Xiaomi: 15 Series Redmi: K70 Pro OPPO: Find N3 Vivo: X90 Pro+ Honor: Magic 5
iOS	iPhone 14	iPhone 14	iPhone 14pro	iPhone 15pro	iPhone 15pro

Phone models examples for the lowest configuration:

	Ferret Series	Otter Series	Raptor/Raptor X	Raptor Pro	Sermoon S1
Android	Samsung: S21 Series Xiaomi: 12 Series OPPO: Find X5 Pro Vivo: X100 Honor: Magic 4	Samsung: S21 Series Xiaomi: 12 Series OPPO: Find N3 Vivo: X90 pro+ Honor: Magic 4	Samsung: S22 Series Xiaomi: 12 Series OPPO: Find N3 Vivo: X90 pro+ Honor: Magic 5	Samsung: S22 Series Xiaomi: 15 Series OPPO: Find N3 Vivo: X90 pro+ Honor: Magic 5	Samsung: S22 Series Xiaomi: 15 Series OPPO: Find N3 Vivo: X90 pro+ Honor: Magic 5
iOS	iPhone 11	iPhone 12	iPhone 13	iPhone 14	iPhone 14

Examples of Unsupported Phone Models:

Due to chip compatibility or system limitations, the following models are currently not supported for connection with the CrealityScan software.

Platform/CPU	Tensor G4	MediaTek Dimensity 9500 / 9400 / 9300 / 8500 / 8300	Huawei Kirin 9020 / 9000S / 8020
Android	Google Pixel 10 Pro Google Pixel 10 Pro Fold Google Pixel 10 Pro XL	OnePlus Ace6 Ultimate Edition vivo X300 vivo X300 Pro vivo X200 Series vivo X100 Series OPPO Find X8 Series Honor Magic8 Pro Air Honor Power 2 Redmi K90 Ultimate Edition Redmi Turbo 5 Redmi Note 15 Pro+ Realme GT Neo6	Mate 70 Series Mate X6 Mate 60 Series Pura 80 Series Nova 14 Ultra

Performance Evaluation Criteria and Detailed Explanations

I. Theoretical Evaluation:

1. Minimum System Requirements

- Definition Interpretation: This is the "threshold" that can barely start and run the scanning software and basic functions. Meeting this configuration means that the software can theoretically be installed and opened, but it is almost impossible to guarantee any smoothness or functional integrity. Therefore, please ensure that your configuration is better than the minimum requirements.

2. Recommended System Requirements

- Definition Interpretation: This is the "ideal configuration" that can ensure the smooth operation of the scanner's core functions. With this configuration, you can enjoy smooth previews, stable frame rates, and efficient data processing, fully leveraging the scanner's various functions. Therefore, please ensure that your device can reach or exceed the recommended configuration on the basis of the minimum requirements.

II. Actual Computer Performance Test within the Software

Definition and Interpretation: This is the most direct and accurate method of judgment. The built-in performance testing tool in the Creality scanning software will comprehensively evaluate your CPU, memory, graphics card, and driver environment, and provide a clear result of the computer performance test.

Operation and Interpretation:

- This test usually runs automatically when you start the software for the first time.

- You can also manually find the "Performance Test" function in the "Settings" menu of the software.
- If the test passes (normal / smooth / very smooth): Congratulations! Your computer meets the basic operation requirements of this software under the current driver and system environment.
- If the test result shows "extremely low": Please pay attention to this result. The software will clearly indicate which hardware (usually the graphics card) does not meet the standard. If the problem persists, it indicates that your hardware configuration indeed needs to be upgraded.
- Frame rate is the direct standard for judging whether the performance meets the requirements. If the maximum frame rate can only reach a few frames, it will seriously affect the scanning effect.
- When encountering faults suspected to be related to computer performance, you can copy the test results from the performance test to the after-sales engineer with one click.

Additional suggestions

External power supply: For laptops, when performing wired scanning, it is essential to connect the power adapter and switch to the "High Performance" power plan. This will lift the system's power consumption restrictions on hardware, allowing the CPU and graphics card to operate at full capacity.